

LOVELY PROFESSIONAL UNIVERSITY

Sample Question Set - Unit 3

Course Code: INT396

Course Title: UNSUPERVISED LEARNING

Course Outcome: CO3

Paper: Sample Multiple Choice Questions - ANSWER KEY

Maximum Marks: 40 x 1 = 40

Unit: 3

Unit Title: Hierarchical and Density-Based Clustering

Unit Mapped RBT: L3 (Apply)

Total Questions: 40

Suggested Duration: 60 Minutes

Answer Grid (Key)

Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
Q1	A	Q11	D	Q21	A	Q31	C
Q2	B	Q12	B	Q22	C	Q32	A
Q3	C	Q13	D	Q23	B	Q33	A
Q4	D	Q14	C	Q24	D	Q34	D
Q5	B	Q15	B	Q25	C	Q35	B
Q6	D	Q16	A	Q26	D	Q36	C
Q7	A	Q17	B	Q27	A	Q37	C
Q8	C	Q18	A	Q28	B	Q38	B
Q9	C	Q19	D	Q29	D	Q39	D
Q10	A	Q20	C	Q30	B	Q40	A

Multiple Choice Questions (40 x 1 = 40 Marks)

Q1. Name the hierarchical approach that begins with every record in one cluster and splits it repeatedly.

- A) Divisive clustering <-- correct
- B) Agglomerative clustering
- C) Density-based clustering
- D) Partitional clustering

Answer: A. Divisive clustering works top down, splitting until each record stands alone. Agglomerative clustering works the other way, from singletons upwards.

Q2. Identify the label a DBSCAN implementation conventionally gives to a noise point.

- A) 0
- B) -1 <-- correct
- C) 1
- D) Null

Answer: B. Clusters are numbered from 0 upwards and noise is marked -1, which keeps the noise points distinguishable from any genuine cluster.

Q3. State the linkage criterion that merges the pair of clusters producing the smallest increase in the total within-cluster variance.

- A) Single linkage
- B) Complete linkage
- C) Ward linkage <-- correct

D) Average linkage

Answer: C. Ward linkage chooses the merge that costs the least in added spread, which is why it tends to produce compact, roughly equal clusters.

Q4. Name the table of pairwise distances that agglomerative clustering has to hold while it runs.

- A) The confusion matrix
- B) The covariance matrix
- C) The transition matrix
- D) The proximity matrix <-- correct

Answer: D. The proximity matrix records the distance between every pair, and it is the reason the memory cost grows with the square of the number of records.

Q5. Identify the plot conventionally used to choose a value of eps for DBSCAN.

- A) The scree plot
- B) The k-distance plot <-- correct
- C) The elbow plot of inertia
- D) The silhouette plot

Answer: B. The distance to each point's k-th nearest neighbour is sorted and plotted; the knee of that curve is read as the value of eps.

Q6. State the number of merges agglomerative clustering performs on n records before a single cluster remains.

- A) n
- B) n + 1

C) $n / 2$

D) $n - 1$ <-- correct

Answer: D. Each merge reduces the number of clusters by one, so going from n singletons to one cluster takes exactly $n - 1$ merges.

Q7. Explain why agglomerative clustering does not require the number of clusters before it runs.

A) It builds the whole hierarchy and is cut afterwards <-- correct

B) It discovers the number from the density of points

C) It always reports exactly two final clusters

D) It requires the number just as k-Means does

Answer: A. The algorithm merges all the way to a single cluster. The number of groups is decided later, by placing a horizontal cut across the dendrogram.

Q8. Distinguish agglomerative clustering from divisive clustering by the direction in which each one builds.

A) Divisive builds upwards, agglomerative builds downwards

B) Both of the two build upwards from single records

C) Agglomerative builds upwards, divisive builds downwards <-- correct

D) Both of the two build downwards from one cluster

Answer: C. Agglomerative starts from singletons and merges; divisive starts from one cluster and splits. Agglomerative is far more commonly used.

Q9. Describe what a horizontal cut placed across a dendrogram produces.

A) A single cluster containing every record

B) A ranking of the records by their height

C) A partition whose size is the number of lines crossed <-- correct

D) A list of the records labelled as noise

Answer: C. Each vertical line the cut crosses is one surviving cluster at that height, so moving the cut up or down changes how many clusters are reported.

Q10. Interpret a dendrogram on which nearly every merge takes place at a similar height.

A) The data shows little clear group structure <-- correct

B) The data holds exactly two natural groups

C) The linkage criterion was computed wrongly

D) The records are duplicates of one another

Answer: A. Distinct groups produce a long gap before the final merges. When every merge costs about the same, no cut is better justified than any other.

Q11. Predict the effect on a DBSCAN result of raising MinPts while eps is held fixed.

A) More core points and less noise

B) The same clusters with new labels

C) One single cluster containing everything

D) Fewer core points and more noise <-- correct

Answer: D. A point now needs more neighbours in the same radius to qualify as a core point, so sparse regions stop forming clusters and are labelled noise.

Q12. Recognize the linkage criterion that measures the distance between two clusters as the distance between their two furthest members.

A) Single linkage

B) Complete linkage <-- correct

C) Average linkage

D) Ward linkage

Answer: B. Complete linkage only permits a merge when even the most distant pair is close, which penalises elongated clusters and favours compact ones.

Q13. Explain why single linkage produces the effect known as chaining.

A) It requires every pair of points to be close

B) It merges the two largest clusters at each step

C) It measures distance from the cluster centres

D) One close pair is enough to join two groups <-- correct

Answer: D. Because the criterion is the nearest pair, a thin trail of intermediate records is enough to bridge two otherwise separate groups into one long cluster.

Q14. Describe what the eps-neighbourhood of a point contains.

A) Every point in the same cluster as it

B) The MinPts points nearest to it

C) Every point lying within distance eps of it <-- correct

D) Every point that is not labelled noise

Answer: C. The eps-neighbourhood is defined purely by the radius. How many points fall inside it is what the MinPts test then counts.

Q15. Explain why the cluster a DBSCAN border point ends up in can depend on the order the records are processed.

A) A border point is assigned entirely at random

B) A border point may lie near two different core points <-- correct

C) A border point is always relabelled as noise

D) A border point changes the value of eps

Answer: B. A border point is reachable from more than one cluster in that case, and implementations give it to whichever cluster claims it first. Core points and noise are not affected.

Q16. Distinguish the memory agglomerative clustering needs from the memory k-Means needs.

A) Agglomerative grows with the square of the records <-- correct

B) k-Means grows with the square of the records

- C) Both of the two grow with the square of the records
 D) Neither of the two depends on the record count

Answer: A. The proximity matrix holds roughly n^2 squared entries, whereas k-Means only has to hold the data and k centres. This is why the two scale so differently.

Q17. Interpret a DBSCAN result that consists of one very large cluster with no noise points at all.

- A) The value of eps has been set too small
 B) The value of eps has been set too large <-- correct
 C) The value of MinPts has been set too large
 D) The dataset contains no genuine structure

Answer: B. With a large enough radius every dense region becomes reachable from every other, so the separate groups merge and nothing is left in a sparse region.

Q18. Predict where two identical records will be joined on a dendrogram.

- A) At a height of zero, before any other merge <-- correct
 B) At the very top, after every other merge
 C) At the average height of the other merges
 D) They are never joined to one another

Answer: A. The distance between them is zero, and agglomerative clustering always merges the closest pair first.

Q19. Solve the following: agglomerative clustering begins with 30 separate records. Determine how many clusters remain after 25 merges.

- A) 6
 B) 25
 C) 29
 D) 5 <-- correct

Answer: D. Each merge removes one cluster, so $30 - 25 = 5$ clusters remain.

Q20. Use single linkage to determine the distance between two clusters whose four inter-cluster pairwise distances are 8, 3, 6 and 5.

- A) 8
 B) 5.5
 C) 3 <-- correct
 D) 6

Answer: C. Single linkage takes the smallest of the cross-cluster distances, which is 3.

Q21. Apply average linkage to two clusters holding 3 and 4 records whose twelve pairwise distances sum to 60.

- A) 5 <-- correct
 B) 12
 C) 7.5
 D) 20

Answer: A. Average linkage is the mean of all cross-cluster pairs: $60 / (3 \times 4) = 60 / 12 = 5$.

Q22. Compute the number of distinct pairwise distances that must be held for 200 records.

- A) 40,000
 B) 20,000
 C) 19,900 <-- correct
 D) 200

Answer: C. The count is $n(n-1)/2 = 200 \times 199 / 2 = 19,900$, which is why the memory cost becomes prohibitive as n grows.

Q23. Apply the DBSCAN definition with MinPts set to 5 to a point that has 6 points inside its eps-neighbourhood, itself included, and determine its status.

- A) A border point
 B) A core point <-- correct
 C) A noise point
 D) It cannot be decided

Answer: B. The neighbourhood count reaches the MinPts threshold, so the point qualifies as a core point and can start or extend a cluster.

Q24. Examine a point that has only 2 neighbours inside its eps-neighbourhood with MinPts set to 5, but which lies inside the eps-neighbourhood of a core point, and determine its status.

- A) A core point
 B) A noise point
 C) A centroid
 D) A border point <-- correct

Answer: D. It is not dense enough to be a core point, but it is reachable from one, so it joins that cluster as a border point rather than being discarded.

Q25. Use a dendrogram cut that crosses four vertical lines to determine the number of clusters reported.

- A) 2
 B) 5
 C) 4 <-- correct
 D) 8

Answer: C. Each vertical line crossed by the cut is one cluster surviving at that height, so four lines give four clusters.

Q26. Compute the number of merges still to be performed when agglomerative clustering of 50 records has reached 12 clusters.

- A) 12
 B) 38
 C) 49
 D) 11 <-- correct

Answer: D. Going from 12 clusters to one takes $12 - 1 = 11$ further merges.

Q27. Apply complete linkage to two clusters whose four inter-cluster pairwise distances are 2, 9, 4 and 7.

- A) 9 <-- correct
- B) 2
- C) 5.5
- D) 4

Answer: A. Complete linkage takes the largest of the cross-cluster distances, which is 9.

Q28. Compare single linkage with Ward linkage on the shape of cluster each one tends to produce.

- A) Single gives compact groups, Ward gives long chains
- B) Single gives long chains, Ward gives compact groups <-- correct
- C) Both of the two give long straggling chains
- D) Both of the two give compact rounded groups

Answer: B. Single linkage merges on the nearest pair and follows thin trails; Ward penalises added variance and therefore keeps clusters tight.

Q29. Use the k-distance plot to determine where the value of eps should be read.

- A) At the very start of the sorted curve
- B) At the very end of the sorted curve
- C) At the midpoint of the record count
- D) At the knee, where the distance begins to rise sharply <-- correct

Answer: D. Below the knee the neighbourhoods are dense and comparable; beyond it the distances climb, which marks where points stop belonging to a dense region.

Q30. Examine a DBSCAN run in which eps has been set to a very small value, and determine the result.

- A) Every record falls into one cluster
- B) Almost every record is labelled as noise <-- correct
- C) The clusters are unusually well separated
- D) The algorithm reports exactly two clusters

Answer: B. Hardly any point gathers MinPts neighbours inside so small a radius, so very few core points exist and the clusters never form.

Q31. Apply DBSCAN to a dataset whose two groups form concentric rings and determine whether it can separate them.

- A) No, because it assumes convex clusters
- B) No, because it needs the number of clusters
- C) Yes, because it follows connected dense regions <-- correct
- D) Yes, but only after the data is standardised

Answer: C. DBSCAN grows a cluster along whatever path stays dense, so a ring is recovered as a single connected region however curved it is.

Q32. Compare the number of parameters that k-Means requires with the number that DBSCAN requires.

- A) k-Means needs one, DBSCAN needs two <-- correct
- B) k-Means needs two, DBSCAN needs one
- C) Both of the two need exactly one
- D) Both of the two need exactly three

Answer: A. k-Means needs k; DBSCAN needs eps and MinPts but not the number of clusters, which it discovers from the density.

Q33. Compute the number of pairwise distances between a cluster holding 4 records and a cluster holding 5 records.

- A) 20 <-- correct
- B) 9
- C) 10
- D) 45

Answer: A. Every record of one is paired with every record of the other: $4 \times 5 = 20$ cross-cluster distances.

Q34. Use the comparison of the three clustering families to select a method for a dataset of irregular shapes and many stray records where the number of groups is unknown.

- A) k-Means
- B) k-Medoids
- C) MiniBatch k-Means
- D) DBSCAN <-- correct

Answer: D. DBSCAN needs no k, recovers shapes of any form and sets the strays aside as noise, which is exactly the combination the brief demands.

Q35. Examine the following and select the correct explanation: agglomerative clustering is run on 500,000 records and the machine runs out of memory before any result appears.

- A) The dendrogram cannot be drawn for so many records
- B) The proximity matrix grows with the square of the records <-- correct
- C) The linkage criterion was chosen incorrectly
- D) The records were not standardised beforehand

Answer: B. Half a million records implies over a hundred billion pairwise distances, which no ordinary machine can hold. A sample or a partitioning method is needed instead.

Q36. Apply Ward linkage and determine what it minimises at each merge.

- A) The distance between the two nearest members
- B) The distance between the two furthest members
- C) The increase in the within-cluster sum of squares <-- correct
- D) The number of records in the merged cluster

Answer: C. Ward evaluates each candidate merge by how much extra spread it introduces and chooses the cheapest, which keeps the resulting clusters tight.

Q37. Demonstrate how the number of clusters is obtained from a completed dendrogram.

- A) Count the number of records at the bottom
- B) Count the total number of merges performed
- C) Place a horizontal cut and count the lines it crosses <-- correct
- D) Read the height of the very last merge

Answer: C. The cut is normally placed in the largest vertical gap between consecutive merges, since a long gap means the next merge joins groups that are far less alike.

Q38. Relate the value of eps to the number of clusters DBSCAN reports.

- A) A larger eps splits clusters into more of them
- B) A larger eps merges separate clusters into fewer <- - correct
- C) The value of eps has no effect on the count
- D) The count is fixed before the algorithm runs

Answer: B. Raising the radius makes more points mutually reachable, so neighbouring dense regions join up and the reported number of clusters falls.

Q39. Use the definition of a core point to determine the status of a point that has one fewer neighbour than MinPts within eps and lies near no core point at all.

- A) A core point
- B) A border point
- C) A cluster centre
- D) A noise point <-- correct

Answer: D. It fails the density test itself and is not reachable from anything that passes it, so it belongs to no cluster and is set aside as noise.

Q40. Compare hierarchical clustering with k-Means on whether repeated runs give the same answer.

- A) Hierarchical is deterministic, k-Means depends on its start <-- correct
- B) k-Means is deterministic, hierarchical depends on its start
- C) Both of the two give a different answer each run
- D) Both of the two are completely deterministic

Answer: A. Given the same data and linkage, agglomerative clustering always produces the same dendrogram. k-Means converges to whichever local minimum its random start leads to.

----- End of Sample Question Set -----